Full recovery on one RTX PRO 6000

This is an observation report for a bounded engineering study. It reports complete public-input recoveries, failures and explicit limits. It does not certify a speedup, establish a largest possible ECDLP size, promote a research hypothesis, or implement the paper's conditional tensor contraction algorithm.

Implementation and accounting

The standalone reduction now certifies a prime group order using a nonzero prime-order point and a unique multiple in the exact integer Hasse interval. Tonelli–Shanks replaces the exhaustive root table. Modular products use u128 when required. Deferred dependency reconstruction removes repeated dense provenance propagation; certificate replay uses sparse relation entries. The original tiny constructor and default 512-column cap remain. Explicit limits admit up to 16384 columns; the driver accepts `p < 2^60` and `r < 2^61`.

GPU inverse-hash base construction and generic meet-in-the-middle relation finding feed CPU elimination and scalar extraction. Public Q is obtained by coordinate lifting; its scalar is neither generated nor supplied. Every success is serialized, replayed in a fresh CPU verifier, compared byte-for-byte across arms where prescribed, and subjected to a scalar-plus-one rejection check.

Each cold solve includes target-coordinate construction, process/GPU startup, curve validation, all base trials, relation collection, elimination, extraction, in-process replay, serialization and teardown. Total audit time also includes fresh replay, corruption rejection and certificate comparison. Fixture construction, compilation and preflight are recorded separately. Shared RunPod is L0; every exported wall_valid flag is false.

Initial ladder: all 44 prescribed executions

Source: `614791252805f64a618a8ce0080e8f4f040c3a8f`. Pod: `ao7xe0bzb6wgz9`, started 2026-10-30T03:19:20UTC, one RTX PRO6000 Blackwell Server Edition. Frozen protocol-v3; seeds 101/202, eight slots, tau=2ceil(r^(1/8))+2, max_rows=2^29, chunk=2^20, max_work=2^30, max_columns=16384, max_trials=16. Solve deadline 300 s; replay 60 s.

There were 40 independently replayed recoveries through 52-bit fields and four 56-bit timeouts. Both GPU invocations and both public targets passed at 52 bits. CPU reference ran only at 20/24 bits and matched exact certificates. The native analyzer passed; controller and worker exited 0; pod deletion was verified. All 13 Rust tests, host/CUDA differentials, reduction/replay, Clippy, memcheck, racecheck and synccheck passed.

Field bits	Cold solve(s)	Oracle(s)	Linear(s)	Relations/attempts
20	0.434163	0.010898	0.000162	57/57
24	0.374702	0.013362	0.000211	71/86
28	0.348764	0.011190	0.000457	105/106
32	0.369914	0.022081	0.000781	128/215
36	0.541954	0.118107	0.002500	183/300
40	0.729476	0.290260	0.006634	267/380
44	1.172951	0.661162	0.016964	384/505
48	9.034036	8.175220	0.047612	495/954
52	64.277743	63.106494	0.143088	743/1050
56	300.009571 timeout	Incomplete	Incomplete	Incomplete

Recorded values for arm u / seed 101, rounded for display; no averaging or pooling. All other executions are retained in rows.csv and the graph. At 52 bits, the four cold solves were 64.059441,64.277743,71.206916,71.240142 s. Oracle time was 62.884962–70.030751 s; linear time was 0.117555–0.143088 s. These stage observations motivated a lookup index. They do not identify hardware counters or establish which kernel instruction dominates.

Every initial-ladder execution



Blue: candidate/repeat; green: baseline/CPU; red: timeout/limit. All points retained.

Presentation rendering of recorded rows; log seconds, all 44 executions, including all four timeouts. Run ao7xe0bzb6wgz9; two fixed targets per field, GPU u/x on 20–56 bits, CPU h on 20/24 bits. A timeout is a limit observation, not a proof of nonexistence or hardware impossibility.

Preserved launch failures and export boundary

Version 1, pod liii98dyd3b7to, failed before compilation because wrapper commands were joined. Version 2, pod t9kc1zn12mr2tc, passed 13 Rust tests and device/preflight checks, then failed to compile the native chart analyzer. Both produced zero ladder observations. Their full archives and deletion receipts are retained. Version 3 corrected the analyzer literal and made instrument compilation use Rust 2021 with warnings denied; the original frozen protocols are unchanged.

The v3 export passes canonical schema, flow and file-hash validation. The protocol-aware checker has three retained coverage findings: it assumes 3 arms×10 fields×2 seeds=60 rows, whereas the frozen nonrectangular plan explicitly contains 44 rows because CPU h runs on two fields only. The native frozen analyzer verifies the actual 44-cell contract. This report does not claim a full canonical protocol-checker pass or scientific validation-gate approval.

Raw archive SHA-256: `b59d596b382be86e1734c8d09eafa70f11e69baa5b08fdb73bc5b67962128dfb`. Archive: `s3://setec-astronomy-artifacts/validation/ecbench-explicit-x/e2e-20261030-v3/results.tgz`. `results-v3/archive-manifest.json` hashes every raw member; `certificate-audit.json` retains each public input, certificate digest, positive replay output and negative-control output. The manifests supplement the executed verifier, not replace it.

Scope and unresolved comparisons

The relation oracle is generic decomposition, bounded by the square-root barrier. No matched rho or fastest external-tool comparison was run here. This is not an implementation of the paper's conditional subexponential tensor backend. The root crypto/ecbench package was not compiled locally because of the user's disk constraint; standalone cloud validation and current-head repository CI are distinct checks.

Primary mathematical sources: the supplied paper §1.2/§3; the repository's order-witness arithmetic; Hasse/point-order notes at https://www.hyperelliptic.org/tanja/teaching/isogeny-school21/csidh-sidh-week-3.pdf; PARI ellcard documentation at https://pari.math.u-bordeaux.fr/dochtml/html-stable/Elliptic_curves.html.

Indexed lookup: complete 56-bit rung

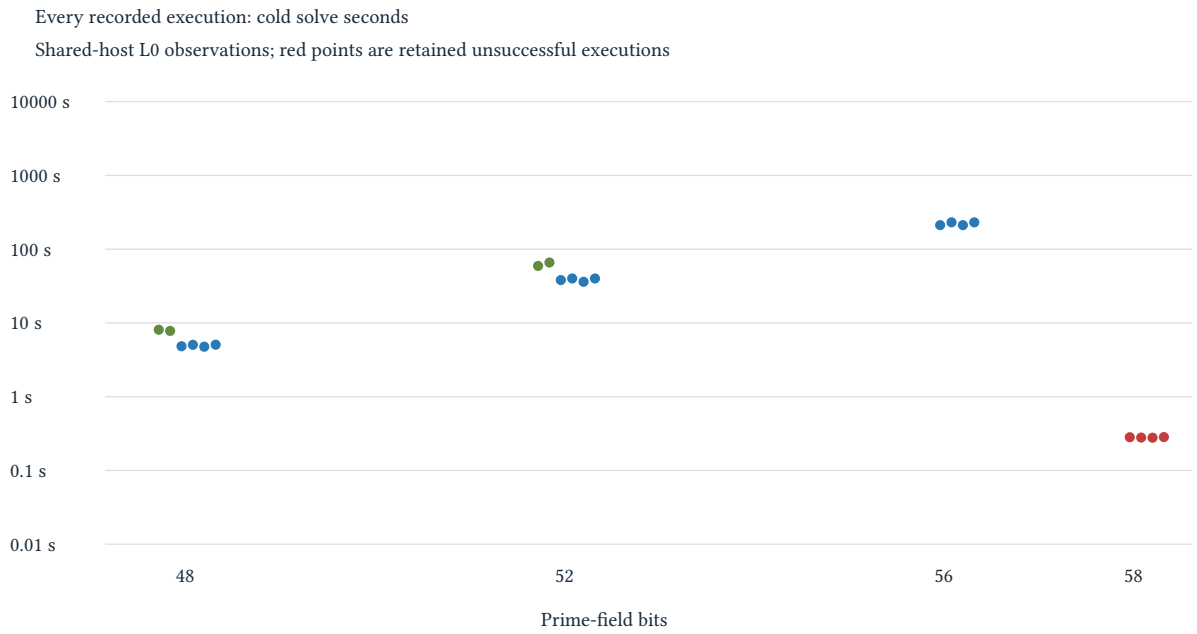
Frozen protocol-v4, source `320694bfff1eaed8e2471e9335d9e7cfa94e8ad4`, pod as4eurugkqwpy4, started 2026-10-30T04:20:00UTC. One RTX PRO6000 Blackwell Server Edition, 97,887 MiB, driver 595.91.07, CUDA 13.2.51. The source adds at most 65,537 u64 prefix offsets to narrow each full-key binary search. Infinity uses the full table; duplicates preserve the lowest original tuple id. Setup and index storage are included in cold timing and GPU admission.

All 20 prescribed executions through 58 bits ran: 16 verified recoveries through 56 bits and four 58-bit work-limit failures. The frozen stopping rule left four potential 60-bit executions unattempted. All successful certificates matched across arms and passed fresh replay and scalar-corruption rejection. The native analyzer, 13 Rust tests, host/CUDA/reduction checks, Clippy and index-specific memcheck/racecheck/synccheck passed. Controller and worker exited 0, and pod deletion was verified.

Bits	Seed	Binary b(s)	Indexed u(s)	Identical x(s)
48	101	8.096602	4.833402	4.762925
48	202	7.799079	5.051183	5.073777
52	101	59.337260	38.096282	36.137007
52	202	65.950374	40.096043	40.053185
56	101	Not prescribed	211.902261	212.031528
56	202	Not prescribed	231.374448	230.939895
58	101/202	Not prescribed	Work limit	Work limit

Recorded cold solve seconds, rounded; each successful cell is one complete recovery plus in-process replay. Additional independent replay/audit costs are separately recorded in rows.csv. No values are pooled across hosts or versions. The seed 101 52-bit u/x cold values differ by 1.959275 s despite almost identical oracle times; this variation is retained, not removed.

At 58 bits, seed 101′s planning cap selected 4,033,512 left rows and 87,620,130,000 right rows; that exceeded max_work=2^30. All four 58-bit failures were explicit admission/work failures, not exhaustive unsuccessful searches. Version 5 separately raises the explicit row/work caps and tests bounded partial-sum reuse; v4 is unchanged.



Blue: candidate/repeat; green: baseline/CPU; red: timeout/limit. All points retained.

All 20 recorded executions, rendered from unchanged rows. Red 58-bit markers are quick admission failures, not fast recoveries. A/A repeats are the exact same indexed service/driver. These L0 timing observations do not certify a speedup.

The export passes schema/flow/file hashes. Its five retained protocol-aware coverage findings arise from the checker's rectangular 30-cell assumption; the frozen contract has at most 24 cells and explicitly stops after the first incomplete rung. Canonical flow calls failed processes completed outcomes: 20 process outcomes comprise 16 recoveries and 4 failures, as coverage.json states.

Raw archive SHA-256: `b93292c25ee0759e94a8f1bfeb89a342a0fd5a4b2701420a760b1e7c948960d1`. Archive and complete certificate/log manifest are linked in results-v4.

Partial-sum reuse: one 58-bit recovery

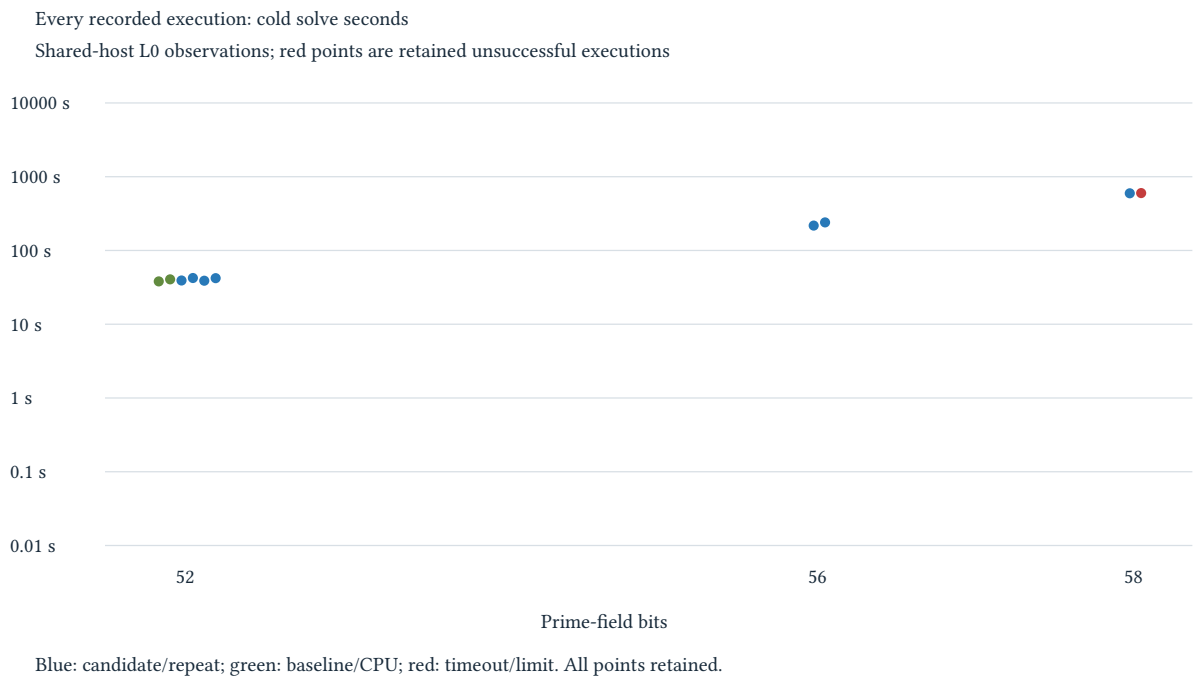
Frozen protocol-v5, measured source `f6338418572c72e1bf97ad24903cd5efd17cf7da`, pod `9fil2lhphdymz0`, started 2026-10-30T04:51:28 UTC. Two contiguous right-side partial sums were cached only when each fit the explicit chunk bound. Their construction and storage were included in setup and admission. The uncached indexed search was the matched baseline; both arms used max_rows=2^30 and max_work=2^31.

Ten executions ran: nine independently replayed recoveries and one timeout. Both targets completed at 56 bits. At 58 bits, seed 101 completed and seed 202 reached its 600-second deadline. The stopping rule left both 60-bit cases unattempted. For v5, the largest complete two-target rung is 56 bits, while the largest individual verified recovery is 58 bits; later panels are separate below.

Bits	Seed	Uncached b (s)	Cached u (s)	Identical x (s)
52	101	38.074392	39.173036	38.890065
52	202	40.590157	42.322563	42.127127
56	101	Not prescribed	217.426942	Not prescribed
56	202	Not prescribed	240.146670	Not prescribed
58	101	Not prescribed	595.862060	Not prescribed
58	202	Not prescribed	600.006124 timeout	Not prescribed

Recorded cold solve seconds, rounded. Cached and A/A executions were slower than their matched uncached baseline at both 52-bit targets. Consequently ordinary builds retain uncached indexed search; caching is explicitly available with `make cuda-cached`. The running panel was never changed. No larger-field matched timing comparison was prescribed.

The successful 58-bit recovery used 1,251 relations from 1,714 attempts. Its recorded oracle time was 593.926630 seconds and linear time 0.725390 seconds; complete execution with additional audit took 596.139162 seconds. The retained table had 605,026,800 left rows and 584,134,200 right tuples, with conservative admission 58,084,287,312 bytes. The other target admitted 583,697,295 left rows and 535,791,186 right tuples, then timed out. Neither 58-bit attempt was refused by memory admission. Admission is not peak VRAM.



All ten executions are shown, including the timeout; two higher cells were unattempted. All 13 Rust tests, host/device/reduction checks, Clippy and partial-cache boundary checks passed, with zero memcheck/synccheck errors and zero racecheck hazards. Both cached and uncached service variants passed preflight. Controller and worker exited zero; pod deletion was verified.

Schema/flow/file verification passes. The five retained protocol-aware coverage findings reflect its rectangular 24-cell assumption versus the explicit 12-cell maximum and stopping rule. The native analyzer verifies the actual ten cells. Raw archive SHA-256: `e5677cb35cc3a34421126fe7d5aad1cac041b1bf8c7a678fd4eca0a3b3196e9b`. Complete custody and replay outputs are in `results-v5` and its linked S3 archive.

Exact compact tables: retained launch failure

The report graphs use the adjacent `render.rs` presentation source with a fixed 0.01–10,000-second logarithmic axis, keeping large results clear of the heading. Original frozen analyzer graphs remain unchanged in each result directory. No measurements are pooled, fitted, omitted or recomputed.

The next variant stores an exact point key and original tuple ID in 16 bytes, instead of the full-coordinate row's 32 bytes. For valid points on a fixed odd-prime curve, x and y parity identify the full point (SEC 1 v2, section 2.3.3: https://www.secg.org/sec1-v2.pdf). The internal key is `2*(x+1)+(y&1)`, reserving zero for infinity. Under `p < 2^62`, both keys and prefix endpoints fit u64. This is exact compression, not a fingerprint. Duplicate keys retain the lowest original tuple ID. Partial caching is off.

V6 source `5a7bfdb97af8941261a2e66a9008231e52f3f23a` ran on pod `oazm8b2kvq0wvz`, created 2026-10-30T05:27:29 UTC. Full-coordinate, cached and compact CUDA builds and recovery checks passed, along with 13 Rust tests, Clippy and full/compact query memcheck, racecheck and synccheck. The first 52-bit baseline solve then failed: the service header still rejected max_rows above 2^30, whereas the frozen panel requested 2^31. The ladder aborted before recording its first row. This was an implementation failure, not a frontier observation. Worker exit was 101, controller exit 1; artifact collection succeeded and deletion was verified.

Raw archive SHA-256: `01b72b36f186beacc5383891fa6a6c540806480c7515dd7949d6ac1737f73db2`. The failure record, first input/error, sanitizer outputs and complete raw member hashes remain in `failed-launch-v6`. The failed protocol is unchanged.

Launch	Build / preflight	Ladder boundary	Recorded rows
v1	Wrapper failed before compilation	Never entered	0
v2	Service checks passed; chart analyzer compilation failed	Never entered	0
v6	All service variants and query sanitizers passed	First service header rejected the requested cap	0

All failed launches are retained; none contributes a successful recovery or an inferred timing advantage. No task GPU was launched until the preceding pod's deletion had been verified.

Frozen protocol-v7 authorizes one new execution after raising the explicit service cap to 2^31, matching the checked u64 planner/device bounds. The cap itself allocates no memory. GPU admission still charges three actual row buffers plus other allocations and preserves 25% free memory. New unit checks cover the upper cap, zero, overflow and chunk bounds. The new wrapper also tests the actual maximum-cap header on both measured services before any ladder data. Inputs, targets, arms, ladder, analyzer and resource limits are unchanged from v6. The cap-fix source is `4014622e18caef9a6b6c89d79653e7633f5aaaff`; v7 outcomes follow separately.

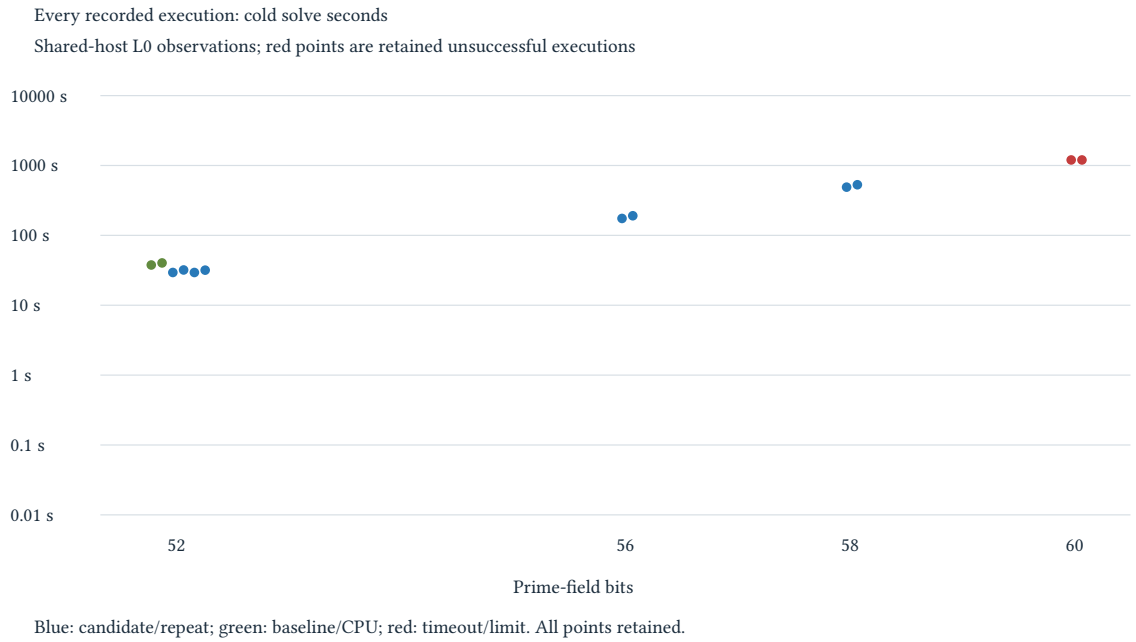
Compact tables: both 58-bit targets recovered

Protocol-v7 ran source `4014622e18caef9a6b6c89d79653e7633f5aaaff` on pod `ex8tm7votw4h4g`, created 2026-10-30T05:37:11 UTC. The device was an RTX PRO 6000 Blackwell Server Edition with 97,887 MiB, driver 595.91.07 and CUDA 13.2.51. All 12 prescribed executions ran: ten independently replayed recoveries and two 60-bit timeouts. Both 58-bit targets completed. The largest complete rung in this panel is therefore 58 bits under its fixed bounds.

Bits	Seed	Full row b (s)	Compact u (s)	Identical x (s)
52	101	37.660576	29.427842	29.424833
52	202	40.318152	31.972281	31.781334
56	101	Not prescribed	174.600283	Not prescribed
56	202	Not prescribed	190.823997	Not prescribed
58	101	Not prescribed	489.753544	Not prescribed
58	202	Not prescribed	530.119796	Not prescribed
60	101	Not prescribed	1200.005057 timeout	Not prescribed
60	202	Not prescribed	1200.006503 timeout	Not prescribed

Recorded cold solve seconds, rounded. Additional fresh CPU verification, corrupt-scalar rejection and certificate comparisons bring the 58-bit total audit times to 490.031165 and 530.390733 seconds. All matched 52-bit certificates are byte-identical. The compact executions have lower observed 52-bit cold times than the full-row baseline; these L0 clocks do not certify a speedup. No matched larger-field timing comparison was prescribed.

The two 58-bit successes used 1,251/1,232 relations from 1,714/1,877 attempts. Their oracle times were 487.892647/528.341356 seconds and linear times 0.727965/0.653090 seconds. The 60-bit tables admitted 1,202,274,400 and 1,025,758,188 left rows, with conservative allocation bounds 57,709,730,784 and 49,236,951,816 bytes. Both later reached the 1,200-second solve limit. Neither was a memory-admission refusal, an exhausted search or a proof that 60-bit recovery is impossible. Allocation bounds are not measured peak VRAM.



Every v7 execution, including both timeouts; no higher cells were omitted. All 13 Rust tests, host/full/cached/compact CUDA checks, Clippy, maximum-cap service headers and full/compact query memcheck, racecheck and synccheck passed. Worker and controller exited zero. Archive collection and pod deletion were verified. Archive SHA-256: `01cd78dbd5cb64410d0c02ba12473964683299f42702a7c5cb46a8f4b7c37891`. The source, raw rows, exact public inputs, complete archive-member hashes, replay outputs and deletion receipt are retained in `results-v7`.

Schema, flow and file hashes pass. The canonical protocol checker retains four rectangular-coverage findings (24 assumed cells versus the frozen 12). The frozen native analyzer and separate strict frontier audit pass the actual contract. The strict audit rejects eight coverage/input/status mutations, including a truncated successful prefix. This does not imply independent scientific validation or erase the canonical findings.

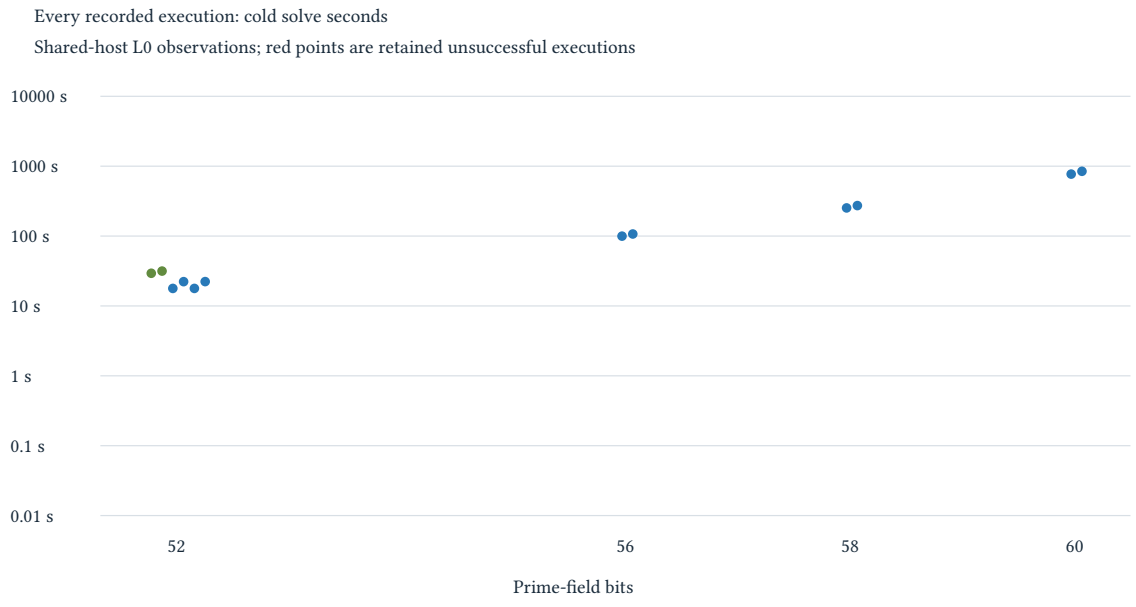
Wider index: both 60-bit targets recovered

Frozen protocol-v8 ran source `bbf0f2f51e73afd7e41174094feb2c56a72d3d9d` on pod `royvvmmehidtwt`, created 2026-10-30T06:50:25 UTC. The RTX PRO 6000 Blackwell Server Edition completed all 12 prescribed executions. Every certificate passed fresh CPU replay and scalar-plus-one rejection; matched 52-bit arms produced identical certificate bytes. The complete frontier of this fixed panel is 60 field bits, with no failed or omitted executions.

Both arms use exact 16-byte rows. The baseline permits up to 2^16 index buckets; the candidate permits 2^24, with at most one bucket per 32 left rows. The largest offsets array is 128 MiB plus eight bytes, included in setup and memory admission. The CPU driver is identical across arms. It includes separately merged arithmetic improvements, so cross-v7/v8 comparisons do not isolate this index change. The wider variant remains explicit opt-in.

Bits	Seed	Index16 b (s)	Wide u (s)	Identical x (s)
52	101	29.344188	17.810696	17.813335
52	202	31.542124	22.262391	22.299082
56	101	Not prescribed	99.731624	Not prescribed
56	202	Not prescribed	107.053471	Not prescribed
58	101	Not prescribed	252.753463	Not prescribed
58	202	Not prescribed	273.055707	Not prescribed
60	101	Not prescribed	770.270416	Not prescribed
60	202	Not prescribed	845.179937	Not prescribed

Recorded cold solve seconds; all clocks remain L0 observations. Complete 60-bit audit totals are 770.303987 and 845.214673 seconds. The two solves use 1,471/1,438 relations from 2,208/2,380 attempts. Oracle time is 768.772528/843.662947 seconds; elimination is 1.108359/1.127755 seconds.



Blue: candidate/repeat; green: baseline/CPU; red: timeout/limit. All points retained.

All 17 Rust tests, host/full/cached/compact device differentials, Clippy, maximum-cap headers and full/compact/wide query memcheck, racecheck and synccheck passed. The wide test checks 2^24 offsets against exact CPU lower bounds, duplicate/parity/infinity cases, and the valid-curve fixtures. Worker/controller exited zero; archive hash and pod deletion were verified. Archive SHA-256: `f8924a3418fd268f0f8492f260b8e7fda66616017e1b60fc84488342a98e7d10`.

Schema/flow/file hashes and strict cell audit pass. Four canonical rectangular coverage findings remain, as for v7; they are not erased by native checks. The CPU driver currently accepts only p < 2^60, so this result is the largest verified rung within that implementation, not the GPU's absolute capacity. A 61/62-bit extension requires a reviewed wider arithmetic/hash path; the next extension is tracked separately under T-906. There is no matched rho or fastest-external-solver result, independent scientific promotion, or GPU implementation of the paper's conditional tensor backend.